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(54) Title: DOSING REGIMENS AND FORMULATIONS OF SUZETRIGINE FOR USE IN THE TREATMENT OF ACUTE AND CHRONIC PAIN

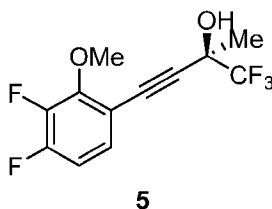
(57) Abstract: Disclosed herein are methods of treating pain, comprising administering suzetrigine or a pharmaceutically acceptable salt thereof, and a process of preparation of suzetrigine.

WO 2025/122953 A1

[0010] In another aspect, the disclosure relates to a pharmaceutical composition comprising: 45-55 wt % of a solid dispersion comprising a polymer and Compound 1; 40-50 wt % of one or more fillers; 2-4 wt % of a disintegrant; and 0.5-1.5 wt % of a lubricant.

[0011] In another aspect, the disclosure relates to a pharmaceutical composition comprising: 45-55 wt % of a solid dispersion comprising a polymer and Compound 1; 42-50 wt % of one or more fillers; 2-4 wt % of a disintegrant; and 0.5-1.5 wt % of a lubricant.

[0012] In another aspect, the disclosure relates to a process for preparing Compound 1, comprising transforming a compound of formula 5:



to Compound 1.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Figure 1 depicts an XRPD pattern of the spray-dried dispersion of Compound 1 of Example 2.

[0014] Figure 2A depicts an XRPD pattern of the SDD tablet composition of Example 2 over the range of about 3° to about 40°2θ.

[0015] Figure 2B depicts an XRPD pattern of the SDD tablet composition of Example 2 over the range of about 14° to about 16°2θ.

[0016] Figure 3A depicts a solid state ¹⁹F NMR spectrum characteristic of the SDD tablet composition of Example 2.

[0017] Figure 3B depicts a solid state ¹³C NMR spectrum characteristic of a powder of the SDD tablet composition of Example 2.

[0018] Figure 4 depicts a graph of mean NPRS score versus time for each treatment group in the study described in Example 5.

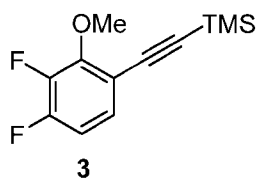
[0019] Figure 5 depicts a graph of mean NPRS score versus time for each treatment group in the study described in Example 6.

DETAILED DESCRIPTION

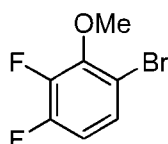
Definitions

[0020] The chemical elements are identified herein in accordance with the Periodic Table of the Elements, CAS version, Handbook of Chemistry and Physics, 75th Ed. Additionally, general principles of organic chemistry are described in "Organic Chemistry," Thomas Sorrell, University Science Books, Sausalito: 1999, and "March's Advanced Organic Chemistry," 5th Ed., Ed.: Smith,

to the compound of formula **5**, optionally wherein the compound of formula **4** is obtained by transforming a compound of formula **3**:

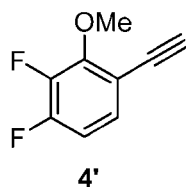


to the compound of formula **4**, optionally wherein the compound of formula **3** is obtained by transforming a compound of formula:

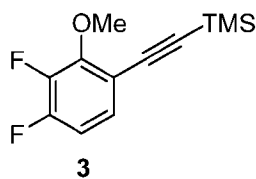


to the compound of formula **3**; or

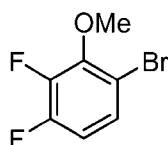
(b) the compound of formula **5** is obtained by transforming a compound of formula **4'**:



to the compound of formula **5**, optionally wherein the compound of formula **4'** is obtained by transforming a compound of formula **3**:

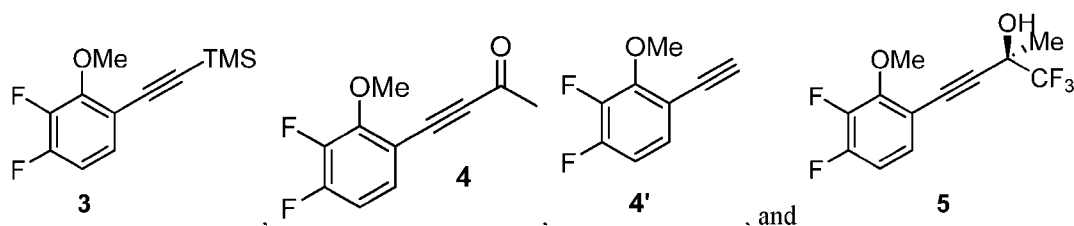


to the compound of formula **4'**, optionally wherein the compound of formula **3** is obtained by transforming a compound of formula:



to the compound of formula **3**.

46. A compound selected from:



or a salt thereof.